

Dietary soy and fats in relation to serum insulin-like growth factor-1 and insulin-like growth factor-binding protein-3 levels in premenopausal Jap...

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Circulating levels of insulin-like growth factor-1 (IGF-1) and insulin-like growth factor-binding protein-3 (IGFBP-3) have each been associated with premenopausal breast cancer risks. We analyzed data from a cross-sectional study of 261 premenopausal Japanese women aged 20-54 yr with adequate nutritional status to evaluate the relationships between concentrations of IGF-1 and IGFBP-3 in serum and dietary intakes of soy, fats and other nutrients. Diet was assessed by a semiquantitative food frequency questionnaire. There was no significant correlation between soy product as well as soy isoflavone intake and serum IGF-1 or IGFBP-3 levels after controlling for age, total energy, percent body fat, and education level. Total fat intake was significantly inversely correlated with serum IGFBP-3 level ($r = -0.13$, $P = 0.04$). The correlations of saturated and monounsaturated fats with serum IGFBP-3 were of borderline significance ($r = -0.12$, $P = 0.06$ and $r = -0.11$, $P = 0.07$, respectively).